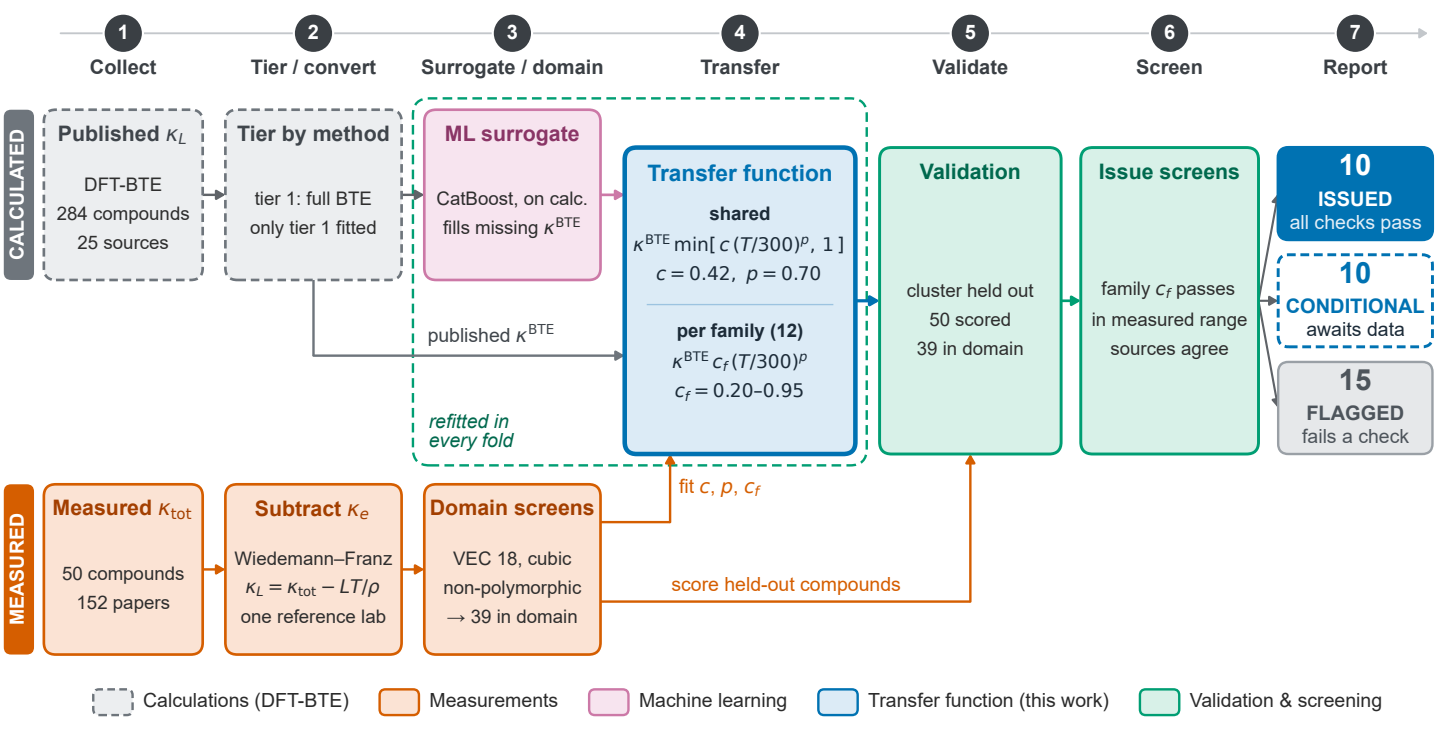

1
Collect
2
Tier / convert
3
Surrogate / domain
4
Transfer
5
Validate
6
Screen
7
Report
CALCULATED
Published κ_L
DFT-BTE
284 compounds
25 sources
Tier by method
tier 1: full BTE
only tier 1 fitted
ML surrogate
CatBoost, on calc.
fills missing κ^{BTE}
Transfer function
shared
$\kappa^{BTE}\min[\,c\,(T/300)^{p},\,1\,]$
$c = 0.42,\ p = 0.70$
per family (12)
$\kappa^{BTE}\,c_f\,(T/300)^{p}$
$c_f = 0.20$–0.95
published κ^{BTE}
refitted in every fold
Validation
cluster held out
50 scored
39 in domain
Issue screens
family c_f passes
in measured range
sources agree
10
ISSUED
all checks pass
10
CONDITIONAL
awaits data
15
FLAGGED
fails a check
MEASURED
Measured κ_{tot}
50 compounds
152 papers
Subtract κ_e
Wiedemann–Franz
$\kappa_L = \kappa_{tot} - LT/\rho$
one reference lab
Domain screens
VEC 18, cubic
non-polymorphic
→ 39 in domain
fit c, p, c_f
score held-out compounds
Calculations (DFT-BTE)
Measurements
Machine learning
Transfer function (this work)
Validation & screening